

Project Initialization and Planning Phase

Date	28-06-2026
Team ID	
Project Title	Emotion Detection & Learning Support Engine
Maximum Marks	5 Marks

Project Proposal (Proposed Solution) report

The proposed solution aims to improve the learning experience of students by combining **Natural Language Processing (NLP)**, **Deep Learning**, and **Generative AI**. The system detects students' emotions from their text-based learning queries and provides personalized academic guidance. By leveraging **BiLSTM**, **BERT**, and **Gemini AI**, the platform delivers fast, accurate, and empathetic responses. The solution reduces manual intervention, enhances learning support, and helps both students and educators identify and address learning challenges effectively.

Project Overview	
Objective	The primary objective of the project is to develop an AI-powered learning assistant that identifies students' emotional states from text input and generates personalized learning guidance using BiLSTM , BERT , and Gemini AI , thereby improving learning outcomes and student engagement.
Scope	The project focuses on building an intelligent learning support platform that performs text preprocessing, emotion detection, confidence score generation, AI-powered response generation, analytics visualization, and interaction logging. It supports students, educators, and academic support teams through a user-friendly Streamlit web application.
Problem Statement	
Description	Students often face difficulties while learning due to confusion, frustration, boredom, or lack of motivation. Traditional learning platforms provide the same content to every learner without understanding their emotional state. Educators also spend considerable time interpreting students' concerns before providing appropriate guidance. This lack of personalized and emotion-aware support negatively impacts the learning experience and academic performance.
Impact	Addressing these challenges will enable faster identification of students' emotional states, provide personalized learning assistance, reduce response time, improve academic engagement, and support educators in delivering timely and empathetic interventions.

Proposed Solution	
Approach	The proposed solution employs Natural Language Processing, Deep Learning , and Generative AI to analyze student text, detect emotions using BiLSTM and BERT models, and generate personalized learning support through Gemini AI . The application is developed using Streamlit , while user interactions and predictions are stored for analytics and continuous improvement.
Key Features	☐ Emotion detection using BiLSTM and BERT models. ☐ AI-generated personalized learning support using Gemini AI. ☐ Mixed emotion detection with confidence scores. ☐ Interactive Streamlit web interface. ☐ Emotion analytics dashboard using Plotly. ☐ CSV logging of user interactions and prediction history. ☐ Comparison of BiLSTM and BERT model performance. ☐ Cloud deployment using Google Cloud Run and Docker.

Resource Requirements

Resource Type	Description	Specification/Allocation
Hardware		
	Computing Resources	Intel Core i3 or above, Google Cloud / Kaggle GPU (T4) for model training
	Memory	Minimum 4 GB RAM (8 GB Recommended)
	Storage	Minimum 2 GB Free Disk Space
Software		
	Frameworks	Streamlit, TensorFlow, Keras, Hugging Face Transformers
	Libraries	NLTK, Plotly, Pandas, NumPy, Scikit-learn, Google Generative AI SDK
	Development Environment	Visual Studio Code, Jupyter Notebook, Anaconda Navigator

Data		
	Data	Emotion-labeled text dataset from Kaggle, CSV format; Trained BiLSTM and BERT model files